KERALA SSLC | MATHEMATICS

SOLIDS

PYQ • 2021 - 2026

1. 2026 **5 MARKS**

(A) The base radius and height of a solid metal cylinder are 8 centimetres and 15 centimetres. By melting it and recasting, how many cones of base radius 6 centimetres and slant height 10 centimetres can be made ?

OR

(B) A toy is in the shape of a square pyramid of base edge 18 centimetres and height 12 centimetres. What is the total cost of painting 500 such toys at 120 rupees per square metre ?

2. 2026 **1 MARK**

Curved surface area of a cone is 60 square centimetres. What is the curved surface area of another cone with same radius but slant height one third of this cone ?

(a) 45 square centimetres (b) 60 square centimetres

(c) 30 square centimetres (d) 20 square centimetres

3. 2025 **5 MARKS**

The base radius of a solid metal cone is 12 centimetres and height 18 centimetres. It is melted and recast into spheres of radius 3 centimetres.

(a) What is the volume of the cone ?

(b) What is the volume of one sphere ?

(c) Find the number of spheres.

4. 2025 **4 MARKS**

A tent in the shape of a square pyramid has length of the base edge 12 metres and height 8 metres.

(a) Find the slant height.

(b) What is the cost of the canvas to make the tent at a rate of 340 rupees per square metre ?

5. 2025 **4 MARKS**

Diameters of two hemispheres are in the ratio 3 : 2.

(a) Write the ratio of their radii.

(b) Write the ratio of their volumes.

(c) If the volume of the first hemisphere is 108 cubic centimetres, what is the volume of the second ?

6. 2024 **5 MARKS**

A vessel (without lid) in the shape of a square pyramid, made from a metallic sheet is of base perimeter 80 centimetres and slant height 26 centimetres.

(a) How many square centimetres of the metallic sheet was needed to make the vessel ?

(b) Calculate the height of the vessel.

(c) What is the capacity of the vessel in litres ?

7. 2024 **4 MARKS**

Diameters of two hemispheres are in the ratio 5 : 3.

(a) Write the ratio of their radii.

(b) Find the ratio of their surface areas.

(c) If the surface area of the first hemisphere is 100 square centimetres, what is the surface area of the other ?

8. 2024 **4 MARKS**

From a circle of radius 15 centimetres, a sector of central angle 120° is cut out and rolled up to make a cone.

(a) What is the slant height of the cone ?

(b) What is the base radius of the cone ?

(c) Calculate the curved surface area of the cone.

9. 2023 **5 MARKS**

There are two cylindrical wooden blocks with diameter 60 centimetres and height 60 centimetres. A largest cone is carved out from one block and a largest sphere from the other.

(a) What is the volume of the cylinder ?

(b) Find the volume of the cone.

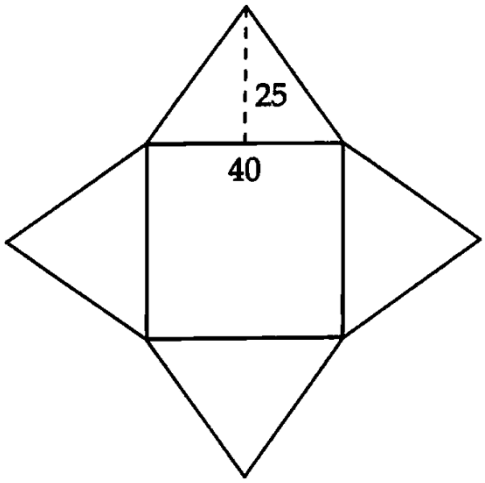
(c) Find the radius of the sphere.

(d) Calculate the volume of the sphere.

(e) Find the ratio of the volumes of the cone and the sphere.

10. 2023 **4 MARKS**

A square pyramid is made by cutting out a paper as in the figure. Side of the square is 40 centimetres. Height of the triangle is 25 centimetres.



(a) What is the slant height of the square pyramid ?

(b) Find the height of the pyramid.

(c) Calculate the volume of the pyramid.

11. 2023 **4 MARKS**

A cone is made by rolling up a semicircle of radius 20 centimetres.

(a) What is the slant height of the cone ?

(b) Find the radius of the cone.

(c) Calculate the curved surface area of the cone.

12. 2022 **6 MARKS**

The radius of a solid metal sphere is 6 centimetres.

(a) Find the volume of the sphere. **(3)**

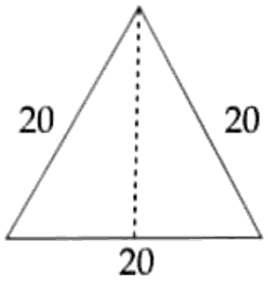
(b) This sphere is melted and recast into a solid cone of radius 6 centimetres. Find the height of the cone. **(3)**

13. 2022 **4 MARKS**

Find the surface area of a cone having base radius 9 centimetres and height 12 centimetres.

14. 2022 **1 MARK**

The lateral faces of a square pyramid are equilateral triangles. If the length of one base edge is 20 centimetres, what will be the measure of its slant height in centimetres ?



(10 ; $10\sqrt{2}$; $10\sqrt{3}$; 20)

15. 2022 **1 MARK**

From the circle of radius 12 centimetres, a sector of central angle 90° is cut out and made into a cone. What is the base radius of this cone ?

16. 2021 **5 MARKS**

(a) Radius of a solid metal cone is 5 centimetres, its slant height is 13 centimetres. Find its height. **(2)**

(b) Find the volume of the cone. **(1)**

(c) It is melted and recast into small cones of radius 1 centimetre and height one centimetre. How many cones will we get ? **(2)**

17. 2021 **5 MARKS**

The diameters of two spheres are in the ratio 1 : 2.

(a) What is the ratio of their radii ? **(1)**

(b) Find the ratio of their surface areas. **(3)**

(c) If the surface area of the first sphere is 10π square centimetres, what is the surface area of the second sphere ? **(1)**